

# Chapter 1 — Introduction to Python

AI Study Assistant — RAG Knowledge Base

## Learning Objectives

Understand what Python is and why it is widely used.

Learn the basic structure of a Python program.

Understand comments, the `print()` function, and simple expressions.

Recognize common uses of Python in software development, data analysis, automation, artificial intelligence, and web development.

### 1. What is Python?

Python is a high-level, general-purpose programming language known for readable syntax and a large standard library. Python programs are commonly used for automation, data processing, web applications, scientific computing, and artificial intelligence.

Python emphasizes readability. Indentation is used to define blocks of code, so consistent indentation is important.

### 2. First Python Program

One of the simplest Python statements is `print()`. It displays a value or message on the screen.

**Example:**

```
print("Hello, Python!")
```

The string inside quotation marks is the value being displayed.

### 3. Comments

Comments are notes written for people reading the program. Python ignores comments when executing the program.

**Single-line comment:**

```
# This is a comment
```

Comments can explain the purpose of a section of code or clarify a calculation.

### 4. Expressions and Operators

An expression combines values and operators to produce a result.

`2 + 3` produces 5.

`10 * 4` produces 40.

Common arithmetic operators include `+`, `-`, `*`, `/`, `//`, `%`, and `**`.

### 5. Python Syntax and Indentation

Python uses indentation to show which statements belong to a block. Unlike languages that use braces for blocks, Python relies on whitespace.

```
if 5 > 2:  
    print("Five is greater")
```

The indented statement belongs to the if block.

## Key Points for Revision

Python is a high-level, general-purpose language.

`print()` is used to display output.

`#` begins a single-line comment.

Indentation is syntactically important in Python.

Arithmetic operators can be used to build expressions.